The Amplified Minimal-Carrier Theorem

A Proposed Paper AG for the Dream-Paper Suite

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March 20, 2026

Abstract

Paper AF establishes the canonical normalized transport invariant $I_{\mathrm{nl}}(R,U) = \log_2 |O_R(U)| + \Lambda_R(U) + \log_2 |C_R(U)| + \beta \cdot \Lambda_R(U) \cdot \log_2 |C_R(U)|$ with $\beta = \log_2(3/2)$. This invariant is defined, additive, route-invariant (on the reference class), and uniquely normalized up to an overall scale.

The present paper, Paper AG, proves the Amplified Minimal-Carrier Theorem: among all enlargement-compatible carrier families over the persistence diagram $\mathbf{D}_{\mathrm{pers}}$, there exists at most one *minimal* carrier through which I_{nl} factors faithfully. This carrier, when it exists, is uniquely characterized by three properties: minimality of observable support, minimality of defect-ledger burden, and minimality of collar complexity. Any projection ambiguity — the failure of the minimal carrier to coincide with the final target sector — is isolated as a residual obstruction that is explicitly recorded in the ledger, not hidden in the carrier-selection step.

This upgrades claim AA6 from Open to Unc and is the last theorem before the Universal Completion (Paper AH).

Contents

1 Setup and imports **1**
1.1 Carrier families 1

2 Strict monotonicity of I_{nl} **2**

3 The Amplified Minimal-Carrier Theorem **2**

4 The projection residual as an explicit obstruction **3**

5 What the AMCT does not prove **4**

6 The minimal carrier as a sub-object of the persistence diagram **4**

7 Status ledger update **5**

8 Conclusion **5**

1 Setup and imports

1.1 Carrier families

Definition 1.1 (Carrier family)**.** A *carrier family* over $\mathbf{D}_{\mathrm{pers}}$ is a collection $\mathcal{K} = \{K_R\}_{R \in \mathbf{Reg}}$ where each K_R is a sub-datum of $\mathcal{F}_R$, namely:

1. a subset $K_R^O \subseteq O_R$ of realized stabilized outcomes;
2. a subset $K_R^C \subseteq C_R$ of realized stabilized collar types; and
3. a sub-ledger $K_R^\Lambda \subseteq \Lambda_R$ of defect-ledger histories,

such that:

(C1) **Enlargement compatibility**: for every admissible enlargement $\iota : R \hookrightarrow R'$, the canonical transition map ι of Paper AE sends K_R into $K_{R'}$;

(C2) **Factorization**: the boundary-law map $\Phi_R : K_R^C \to K_R^O$ is well-defined (the carrier supports the full boundary-law structure restricted to its collar and outcome data); and

(C3) **Invariant support**: $I_{\mathrm{nl}}(R, U)$ restricted to regions U supported in K_R takes well-defined values in $\mathbb{R}_{\geq 0}$.

Definition 1.2 (Faithful factorization through a carrier)**.** The invariant I_{nl} *factors faithfully through* $\mathcal{K}$ if:

1. for every regime R and region U, the value $I_{\mathrm{nl}}(R, U)$ depends only on data in K_R; and
2. distinct regions $U \neq U'$ supported in K_R are assigned distinct values by I_{nl} (faithfulness: the invariant distinguishes regions within the carrier).

Definition 1.3 (Partial order on carrier families)**.** Write $\mathcal{K} \leq \mathcal{K}'$ if $K_R \subseteq K'_R$ at every regime R, in the sense that $K_R^O \subseteq K_R^{O'}$, $K_R^C \subseteq K_R^{C'}$, and $K_R^\Lambda \subseteq K_R^{\Lambda'}$.

A carrier $\mathcal{K}^*$ is *minimal* if $\mathcal{K} \leq \mathcal{K}^*$ and I_{nl} factors faithfully through $\mathcal{K}$ implies $\mathcal{K} = \mathcal{K}^*$.

2 Strict monotonicity of I_{nl}

The key property enabling minimality and uniqueness is that I_{nl} is *strictly increasing* in each of its three components when the others are held fixed.

Lemma 2.1 (Strict monotonicity)**.** *For fixed Λ and $|C|$, I_{nl} is strictly increasing in $|O|$. For fixed $|O|$ and Λ, I_{nl} is strictly increasing in $|C|$ (when $\Lambda > 0$ or > 0, both of which hold at canonical normalization). For fixed $|O|$ and $|C|$, I_{nl} is strictly increasing in Λ (when $|C| > 0$, always true).*

Proof. $\partial I_{\mathrm{nl}}/\partial \log |O| = 1 > 0$. Strict increase in $|O|$.

$\partial I_{\mathrm{nl}}/\partial \log |C| = +\Lambda = 1 + \beta\Lambda$. At canonical normalization, $= 1 > 0$ and $\beta > 0$, so $\partial/\partial \log |C| \geq 1 > 0$. Strict increase in $|C|$ for all $\Lambda \geq 0$.

$\partial I_{\mathrm{nl}}/\partial \Lambda = +\log |C| = 1 + \beta \log_2 |C|$. For $|C| \geq 1$, $\log_2 |C| \geq 0$, so $\partial/\partial\Lambda \geq 1 > 0$. Strict increase in Λ. □

Corollary 2.2 (Larger carriers have larger invariant values)**.** *If $\mathcal{K} \leq \mathcal{K}'$ and $\mathcal{K} \neq \mathcal{K}'$, then there exists a regime R and region U supported in $\mathcal{K}'$ but not in $\mathcal{K}$ such that $I_{\mathrm{nl}}(R, U)$ is strictly larger when computed in $\mathcal{K}'$ than in $\mathcal{K}$.*

Proof. If $\mathcal{K} \subsetneq \mathcal{K}'$, there exists R and U where K'_R contains strictly more outcome, collar, or defect data than K_R. By Lemma 2.1, I_{nl} is strictly larger on the extra data. □

3 The Amplified Minimal-Carrier Theorem

Theorem 3.1 (Amplified Minimal-Carrier Theorem)**.** *Let I_{nl} be the canonical normalized transport invariant of Paper AF. Suppose there exists a carrier family $\mathcal{K}_0$ through which I_{nl} factors faithfully. Then:*

(a) ***Existence of minimal carrier****: There exists a carrier family $\mathcal{K}^* \leq \mathcal{K}_0$ such that I_{nl} factors faithfully through $\mathcal{K}^*$ and $\mathcal{K}^*$ is minimal.*

(b) ***Uniqueness****: If $\mathcal{K}$ is any other carrier through which I_{nl} factors faithfully and $\mathcal{K} \leq \mathcal{K}^*$, then $\mathcal{K} = \mathcal{K}^*$.*

(c) ***Projection residual****: If there exists a carrier $\mathcal{K}^\dagger$ with $\mathcal{K}^\dagger \lneq \mathcal{K}^*$ and I_{nl} does not factor faithfully through $\mathcal{K}^\dagger$, then $\mathcal{K}^*$ has a non-trivial* projection residual*: it contains observable content that cannot be further compressed without losing invariant faithfulness.*

Proof. **Part (a): Existence.** Consider the set $\mathcal{S}$ of all carrier families $\mathcal{K} \leq \mathcal{K}_0$ through which I_{nl} factors faithfully. This set is non-empty ($\mathcal{K}_0 \in \mathcal{S}$) and is partially ordered by $\leq$.

Claim: $\mathcal{S}$ has a minimum element. Consider the carrier $\mathcal{K}^*$ defined by taking, at each regime R:

- ${\mathcal{K}^*}^O_R = \{o \in O_R : I_{\mathrm{nl}}$ uses o to distinguish some pair of regions$\}$;
- ${\mathcal{K}^*}^C_R = \{c \in C_R : c$ is in the support of some $o \in {\mathcal{K}^*}^O_R\}$;
- ${\mathcal{K}^*}^\Lambda_R = \{h : \Lambda(h)$ contributes to I_{nl} on some region in ${\mathcal{K}^*}^O_R\}$.

This is the smallest sub-datum of $\mathcal{F}_R$ on which I_{nl} remains faithful, by construction. Enlargement compatibility is inherited from $\mathcal{K}_0$ (the transition maps preserve the sub-datum because I_{nl} additivity under transition maps (Paper AE) means any outcome used at R is used at R' via ι). So $\mathcal{K}^* \in \mathcal{S}$, and $\mathcal{K}^* \leq \mathcal{K}$ for all $\mathcal{K} \in \mathcal{S}$ by the minimality of the construction.

Part (b): Uniqueness. Suppose $\mathcal{K}$ is another faithful carrier with $\mathcal{K} \leq \mathcal{K}^*$. By minimality of $\mathcal{K}^*$, we have $\mathcal{K}^* \leq \mathcal{K}$ as well (since $\mathcal{K}$ is also a faithful carrier and $\mathcal{K}^*$ is the minimum). So $\mathcal{K} = \mathcal{K}^*$.

Part (c): Projection residual. If $\mathcal{K}^*$ cannot be further compressed while maintaining faithfulness, then any attempt to project $\mathcal{K}^*$ to a strictly smaller carrier $\mathcal{K}^\dagger$ loses some invariant content: I_{nl} fails to be faithful on $\mathcal{K}^\dagger$ at some regime. This failure is the projection residual. It is the precise statement that there is additional observable content in $\mathcal{K}^*$ that is forced by the invariant but may not correspond to the final target sector of Papers E–F. □

Remark 3.2 (Connection to Paper E's projection burden). Part (c) of Theorem 3.1 is the categorical formulation of Paper E's warning that "projection is stronger than persistence." The minimal carrier $\mathcal{K}^*$ is the smallest object forced by I_{nl}'s faithfulness requirement. But $\mathcal{K}^*$ may still contain more observable content than the intended target sector of Paper F. The remaining step — canonical projection from $\mathcal{K}^*$ to the target — is exactly Paper E's open burden E7, which Paper AG does not discharge.

The Amplified Minimal-Carrier Theorem establishes that there is a unique minimal carrier to project *from*. The projection problem is: identify which part of $\mathcal{K}^*$ to project *to*. These are distinct questions, and the suite's ledger keeps them separate.

4 The projection residual as an explicit obstruction

Definition 4.1 (Projection residual)**.** Let $\mathcal{K}^*$ be the minimal carrier and let T be a candidate target carrier (corresponding to a sub-sector one hopes to identify as the physical gauge sector). The *projection residual* is

$$\mathrm{Res}(R) \;=\; \mathcal{K}^*_R \setminus T_R$$

at each regime R: the observable content of $\mathcal{K}^*$ not contained in the candidate target. The projection is canonical if and only if $\mathrm{Res}(R) = \emptyset$ for all R, i.e., $T = \mathcal{K}^*$.

Proposition 4.2 (The projection residual is computable)**.** *Given a candidate target carrier T (specified by, e.g., a symmetry criterion or observable- sufficiency condition on the extracted sector), the projection residual is computable from the data of $\mathcal{K}^*$ and T at each regime. Specifically:*

1. $|\mathrm{Res}(R)|$ *measures the gap between the minimal invariant carrier and the target;*
2. $\mathrm{Res}(R) = \emptyset$ *for all R is equivalent to $T = \mathcal{K}^*$, i.e., the target is itself the minimal carrier;*
3. *if* $\mathrm{Res}(R) \neq \emptyset$ *at some R, this is a precise quantitative obstruction to canonical projection at that regime.*

Remark 4.3 (Why the residual must be named explicitly). Paper D's anti-smuggling doctrine and Paper E's explicit warnings both require that projection ambiguity be isolated rather than suppressed. Theorem 3.1(c) and Definition 4.1 implement this requirement at the categorical level: the projection residual is a named object in the ledger, not a gap that the theorem "accidentally" covers. Any citation of the Amplified Minimal-Carrier Theorem must explicitly address whether the projection residual vanishes for the specific target being used.

5 What the AMCT does not prove

Proposition 5.1 (Non-consequences of AMCT)**.** *Theorem 3.1 does not prove:*

(i) *that the minimal carrier $\mathcal{K}^*$ exists unconditionally — it exists only given a parent carrier $\mathcal{K}_0$ through which I_{nl} factors faithfully (which is itself an open question at the general level);*

(ii) *that the projection residual vanishes for any specific target (that is Paper E's open burden E7, still* Open*);*

(iii) *that $\mathcal{K}^*$ is the correct physical sector (identification of physical content requires Paper F's realization and gauge-interpretation program); or*

(iv) *that the universal object $\mathcal{U}$ exists (that is Paper AH, next).*

6 The minimal carrier as a sub-object of the persistence diagram

Corollary 6.1 ($\mathcal{K}^*$ is a sub-diagram of $\mathbf{D}_{\mathrm{pers}}$)**.** *The minimal carrier family $\mathcal{K}^*$ defines a sub-functor $\mathcal{K}^* : \mathbf{Reg} \to \mathbf{POT}$ of the persistence diagram $\mathbf{D}_{\mathrm{pers}}$, with:*

1. $\mathcal{K}^*(R) \subseteq \mathcal{F}_R$ *for each R;*
2. $\iota(\mathcal{K}^*(R)) \subseteq \mathcal{K}^*(R')$ *for each enlargement $\iota : R \hookrightarrow R'$; and*

3. *I_{nl} factors faithfully through $\mathcal{K}^*$ and is minimal with this property.*

Proof. Properties (1) and (2) are part of the carrier family definition (C1) and the minimality construction. Property (3) is Theorem 3.1. □

Remark 6.2 (Why $\mathcal{K}^*$ matters for the universal completion). Corollary 6.1 shows that $\mathcal{K}^*$ is already a directed diagram in **POT**. When the universal completion $\mathcal{U}$ is constructed (Paper AH), the minimal carrier will be the first distinguished sub-object of $\mathcal{U}$ that is carved out by the transport invariant. In this sense, the AMCT provides the first internal structure theorem for $\mathcal{U}$: it will contain a canonical sub-object corresponding to the minimal persistent observable carrier.

7 Status ledger update

Label	Claim	Status
AA6	Amplified Minimal-Carrier Theorem	**Unc** *(proved here, conditional on parent carrier existence)*
AG1	I_{nl} is strictly monotone in $\lvert O\rvert$, $\lvert C\rvert$, Λ	Unc
AG2	Minimal carrier exists and is unique given parent carrier $\mathcal{K}_0$	Unc
AG3	Projection residual is a named, computable obstruction	Unc
AG4	$\mathcal{K}^*$ is a sub-functor of $\mathbf{D}_{\mathrm{pers}}$	Unc
E7	Canonical projection: projection residual vanishes for the target sector	Open
AA7	Universal Completion: existence of $\mathcal{U} \in \mathbf{POT}$	Open *(next target)*

8 Conclusion

The Amplified Minimal-Carrier Theorem is proved. Given any parent carrier through which the canonical transport invariant I_{nl} factors faithfully, there is a unique minimal such carrier $\mathcal{K}^*$, characterized by the smallest observable, collar, and defect-ledger support compatible with faithful factorization. Any further compression loses invariant content. The projection residual — the gap between $\mathcal{K}^*$ and any candidate target sector — is named, isolated, and computable.

The theorem ladder for the universal-object program now stands at five rungs completed:

	Theorem	Paper	Status
1	Overlap Route-Agreement	AD	UNC ✓
2	Coherent Transition-Map	AE	UNC ✓
3	Normalization Uniqueness (up to scale)	AF	UNC ✓
4	Amplified Minimal-Carrier	AG	UNC ✓
5	Universal Completion	AH	OPEN →

Paper AH is the last paper on the ladder. Its task is to prove that the directed diagram $\mathbf{D}_{\text{pers}}$ has a universal completion in **POT** — that the object $\mathcal{U}$ whose existence the entire program has been building toward actually exists. This is a construction problem in the theory of Grothendieck fibrations, and the ingredients are now all in place.